

Data Structure Assignment - 1

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Q1. Write a program to insert the element at the last and first positions in an array.

(a) If an array is empty.

(b) If the array already has 5 elements.

2. Write a program to delete the element at the last and first positions in an array.

(a) If an array is empty.

(b) If the array already has 5 elements.

Both the questions are in the same file i.e. Q1_2.c

```
#include <stdio.h>

#define MAX_SIZE 100

void insertFirst(int arr[], int *size, int num) {
    for (int i = *size; i > 0; i--) {
        arr[i] = arr[i - 1];
    }
    arr[0] = num;
    (*size)++;
}

void insertLast(int arr[], int *size, int num) {
    arr[*size] = num;
    (*size)++;
}

void deleteFirst(int arr[], int *size) {
    if (*size == 0) {
        printf("Array is empty, nothing to delete.\n");
        return;
    }
    for (int i = 0; i < *size - 1; i++) {
        arr[i] = arr[i + 1];
    }
}
```

```

        (*size)--; // Update the size of the array
        printf("Element at the first position deleted.\n");
    }

void deleteLast(int arr[], int *size) {
    if (*size == 0) {
        printf("Array is empty, nothing to delete.\n");
        return;
    }
    (*size)--; // Update the size of the array
    printf("Element at the last position deleted.\n");
}

int main() {
    int arr[MAX_SIZE];
    int size = 0;
    int num, baseSize, choice;

    printf("Enter the number of initial elements in the array (0 or 5):
");
    scanf("%d", &baseSize);

    if (baseSize != 0 && baseSize != 5) {
        printf("Invalid initial size. Please enter either 0 or 5.\n");
        return -1;
    }

    if (baseSize > 0) {
        printf("Enter %d initial elements: ", baseSize);
        for (int i = 0; i < baseSize; i++) {
            scanf("%d", &arr[i]);
        }
        size = baseSize;
    }

    while (1) {
        printf("\nChoose the operation:\n");

```

```

printf("1. Insert at the first position\n");
printf("2. Insert at the last position\n");
printf("3. Delete at the first position\n");
printf("4. Delete at the last position\n");
printf("5. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);

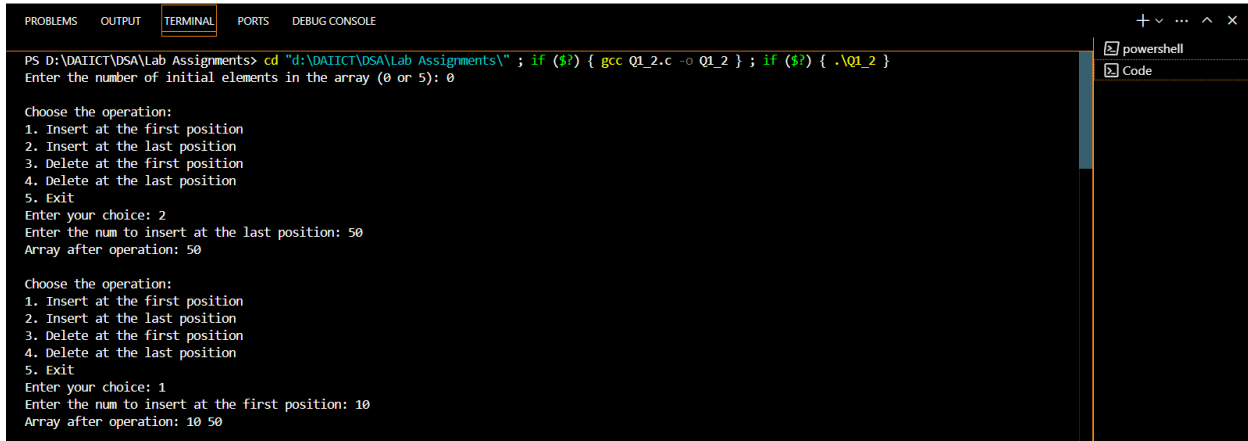
switch (choice) {
    case 1:
        printf("Enter the num to insert at the first position: ");
        scanf("%d", &num);
        insertFirst(arr, &size, num);
        break;
    case 2:
        printf("Enter the num to insert at the last position: ");
        scanf("%d", &num);
        insertLast(arr, &size, num);
        break;
    case 3:
        deleteFirst(arr, &size);
        break;
    case 4:
        deleteLast(arr, &size);
        break;
    case 5:
        printf("Exiting...\n");
        return 0;
    default:
        printf("Invalid choice. Please choose either 1, 2, or
3.\n");
}

printf("Array after operation: ");
for (int i = 0; i < size; i++) {
    printf("%d ", arr[i]);
}
printf("\n");
}

```

```
    return 0;
}
```

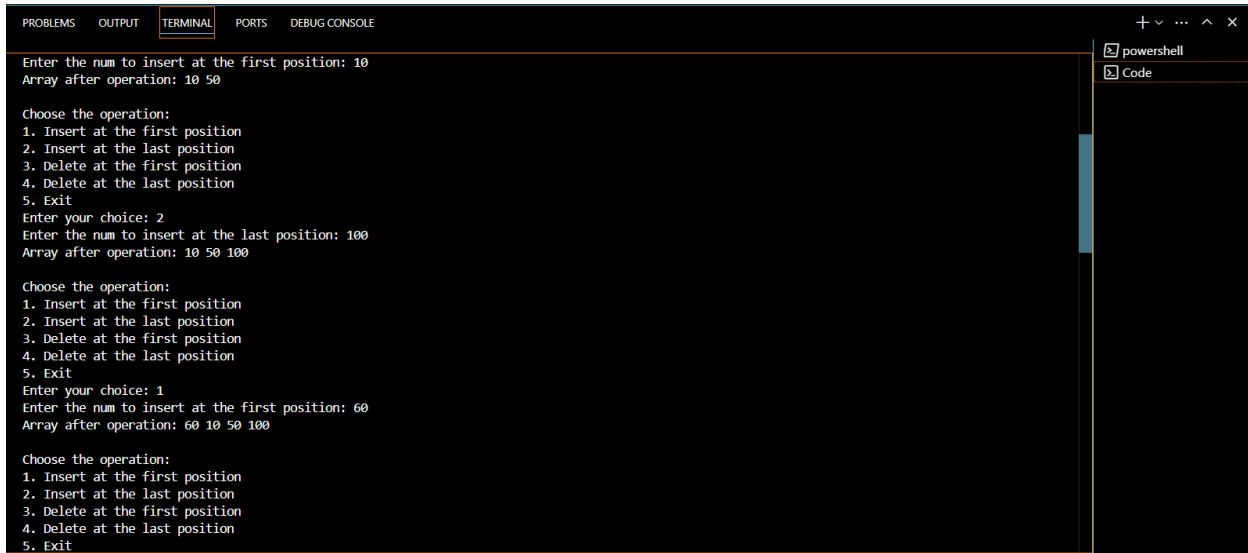
-> Output



```
PS D:\DAIICT\DSA\Lab Assignments> cd "d:\DAIICT\DSA\Lab Assignments\" ; if ($?) { gcc Q1_2.c -o Q1_2 } ; if ($?) { .\Q1_2 }
Enter the number of initial elements in the array (0 or 5): 0

Choose the operation:
1. Insert at the first position
2. Insert at the last position
3. Delete at the first position
4. Delete at the last position
5. Exit
Enter your choice: 2
Enter the num to insert at the last position: 50
Array after operation: 50

Choose the operation:
1. Insert at the first position
2. Insert at the last position
3. Delete at the first position
4. Delete at the last position
5. Exit
Enter your choice: 1
Enter the num to insert at the first position: 10
Array after operation: 10 50
```



```
Enter the num to insert at the first position: 10
Array after operation: 10 50

Choose the operation:
1. Insert at the first position
2. Insert at the last position
3. Delete at the first position
4. Delete at the last position
5. Exit
Enter your choice: 2
Enter the num to insert at the last position: 100
Array after operation: 10 50 100

Choose the operation:
1. Insert at the first position
2. Insert at the last position
3. Delete at the first position
4. Delete at the last position
5. Exit
Enter your choice: 1
Enter the num to insert at the first position: 60
Array after operation: 60 10 50 100

Choose the operation:
1. Insert at the first position
2. Insert at the last position
3. Delete at the first position
4. Delete at the last position
5. Exit
```

```
Choose the operation:
1. Insert at the first position
2. Insert at the last position
3. Delete at the first position
4. Delete at the last position
5. Exit
Enter your choice: 1
Enter the num to insert at the first position: 44
Array after operation: 44 60 10 50 100
```

```
Choose the operation:
1. Insert at the first position
2. Insert at the last position
3. Delete at the first position
4. Delete at the last position
5. Exit
Enter your choice: 3
Element at the first position deleted.
Array after operation: 60 10 50 100
```

```
Choose the operation:
1. Insert at the first position
2. Insert at the last position
3. Delete at the first position
4. Delete at the last position
5. Exit
Enter your choice: 4
Element at the last position deleted.
Array after operation: 60 10 50
```

```
1. Insert at the first position
2. Insert at the last position
3. Delete at the first position
4. Delete at the last position
5. Exit
Enter your choice: 4
Element at the last position deleted.
Array after operation: 60 10
```

```
Choose the operation:
1. Insert at the first position
2. Insert at the last position
3. Delete at the first position
4. Delete at the last position
5. Exit
Enter your choice: 4
Element at the last position deleted.
Array after operation: 60
```

```
Choose the operation:
1. Insert at the first position
2. Insert at the last position
3. Delete at the first position
4. Delete at the last position
5. Exit
Enter your choice: 4
Element at the last position deleted.
Array after operation:
```

```
Choose the operation:
1. Insert at the first position
2. Insert at the last position
3. Delete at the first position
4. Delete at the last position
5. Exit
Enter your choice: 4
Element at the last position deleted.
Array after operation:
```

```
Choose the operation:
1. Insert at the first position
2. Insert at the last position
3. Delete at the first position
4. Delete at the last position
5. Exit
Enter your choice: 4
Array is empty, nothing to delete.
Array after operation:
```

```
Choose the operation:
1. Insert at the first position
2. Insert at the last position
3. Delete at the first position
4. Delete at the last position
5. Exit
Enter your choice: 5
Exiting...
PS D:\DATICT\DSA\Lab Assignments>
```

powershell
Code

powershell
Code

powershell
Code

3. Write a program to find the minimum and maximum elements in an array.
4. Write a program to find duplicate elements in an array in $O(n)$ times.
5. Write a program to arrange the array in increasing order.
6. Write a program to find the largest and second-largest elements in an array.
7. Write a program to reverse an array.

The above questions from 3 to 7 are done in another file Q3_7.c

Comments are attached with the logic of the code to specify which ans is of which question.

```
#include<stdio.h>
#include<conio.h>

int main() {

    int arr[] = {5,10,50,84,200,1,6,99,65,50,1,99};
    int max,min,tmp,j=0;

    int n = sizeof(arr) / sizeof(arr[0]);
    max = arr[0];
    min = arr[0];
    int rev[n];

    //Original Array
    for(int i=0;i<n;i++){
        printf("%d \t",arr[i]);
    }

    //Q-7 Reverse an array
    for(int i=n-1;i>=0;i--){
        rev[j] = arr[i];
        j++;
    }
    printf("\nREVERSE\t");
    for(int i=0;i<n;i++){

        printf("%d ",rev[i]);
    }

    //Q-3 Minimum and maximum in an array
```

```

for(int i=0;i<n;i++){
    if(max<arr[i]){
        max = arr[i];
    }
    if(min > arr[i]){
        min = arr[i];
    }
}

printf("\n%d is maximum\n",max);
printf("%d is minimum",min);

```

//Q-4 Write a program to find duplicate elements in an array in O(n) times.

```

printf("\n-----ARRAY ELEMENTS-----\n");
for(int i = 0; i < n; i++)
{
    printf("%d ",arr[i]);
}
printf("\n-----DUPLICATE ELEMENTS-----\n");
for(int i = 0; i < n-1; i++)
{
    for(j = i+1; j < n; j++)
    {
        if(arr[i] == arr[j])
        {
            printf("%d ",arr[i]);
            break;
        }
    }
}

```

//Q-5 Array in increasing order

```

for(int i=0 ; i<n ; i++){
    for(int j=i+1;j<n;j++){
        if(arr[i]>arr[j]){
            tmp=arr[i];
            arr[i] = arr[j];
            arr[j] = tmp;
        }
    }
}

```

```

    }

}

printf("\nIncreasing order array \t");
for(int i=0;i<n;i++){

    printf("%d ",arr[i]);

}

//Q-6 Largest and the second largest Value of an array
printf("\nLargest No. = %d",arr[n-1]);
printf("\nSecond Largest No. = %d",arr[n-2]);

return 0;
}

```

-> Output

```

PROBLEMS OUTPUT TERMINAL PORTS DEBUG CONSOLE
PS D:\DAIICT\DSA\Lab Assignments>
cd "d:\DAIICT\DSA\Lab Assignments\" ; if ($?) { gcc Q3_7.c -o Q3_7 } ; if ($?) { .\Q3_7 }
5 10 50 84 200 1 6 99 65 50 1 99
REVERSE 99 1 50 65 99 6 1 200 84 50 10 5
200 is maximum
1 is minimum
-----ARRAY ELEMENTS-----
5 10 50 84 200 1 6 99 65 50 1 99
-----DUPLICATE ELEMENTS-----
50 1 99
Increasing order array 1 1 5 6 10 50 50 65 84 99 99 200
Largest No. = 200
Second Largest No. = 99
PS D:\DAIICT\DSA\Lab Assignments>

```